

A Packing of 11 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.045870. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.020672, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.045870 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.020672, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 11/10 = 3.046154$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N11.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
152293539903063950765239548909091581340921023837692464624810198238772587
835640777261945639327108778090452549228077253040339972835751476491796785
9374748959371485413337476302859481025490937240007
```

denominator:

```
500000000000875651133053866840606107696758874002459095358461822898314294
439936028675466537382161463700719506891805162010945978039967615822462261
963045849042228672757793850023440742626408846840
```

B Coordinates

11 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N11.txt`.

```
8039710963 -143698470904 82929355323 103958902306 -130525990421 -161271536920 -8686272174 -31527872098162470588812 -81332918191 231577042023 61352546137
159634243012 271604731084 315590014000 217630929896441905897106 4308170459 486138132109 -85389706135 -61873148420 -462909570308 331328489379 158858741276
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478306765923 112000595080 210115426515 -742594311663801445194 335550611045 345000336872 122857757864 469174846620 157633172649 -23105383370 664639813182
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